

Problem 1: Environment for Cat and Mouse

- Create a custom environment for the Cat and Mouse game (https://gymnasium.farama.org/introduction/create_custom_env/)

Problem 2: Parallel Environments for REINFORCE

- Download the **implementation for policy gradients** from GRIPS!
- Extend the implementation so that it runs N independent environments at each time step!
- Evaluate your solution for the Cat and Mouse game!

Problem 3: The A2C Algorithm

- Download Lapan's **implementation** of the A2C algorithm!
- Use your solution from task 1 to substitute the out-of-date `ptan` library with your own implementation to run several environments in parallel.
- Evaluate your solution for the Cat and Mouse game!

Problem 4: The DQN Algorithm

- Implement DQN for the Cat and Mouse game!